

HVE-CFO Architecture Proposal

Optimized Hermes CFO profile on the approved dual-backend Spark stack
Ollama three-model hot stack · isolated CFO profile · no gpt-oss:120b

DECISION

Stand up hve-cfo on the approved Ollama hot stack.

KEEP

qwen3.8-hermes:27b-128k + 2B deriver + nomic-embed-text.

CONSTRAINT

Share GB10 with hve-coder vLLM at 0.45. No fourth hot LLM.

GAIN

Family-business books, tax RAG, treasury — private, local, cited.

1. Purpose and decision

This note records the proposed inference layout for the **hve-cfo** Hermes profile on the NVIDIA DGX Spark. It is the CFO companion to *HVE-Coder Dual-Backend Architecture* and incorporates the approved decisions in *2026-09-02-hve-coder-dual-backend-decisions-v1.0.md* (Luna, Head Architect and CTO). Those decisions are binding constraints, not suggestions.

The daily HVE agent (**hanshermesagent**) stays on Ollama with the exact tag **qwen3.8-hermes:27b-128k**. The HVE-coder profile evaluates NVIDIA Nemotron 3.5 Lightning on a native vLLM service at gpu-memory-utilization **0.45**. HVE-CFO does not get a third engine and does not load **gpt-oss:120b** or any other large resident model. It is a specialized profile on the already-approved three-model Ollama hot stack, with a dedicated home, workspace, skills, RAG corpus, and deterministic calculators.

Decision. Create hve-cfo as a third isolated Hermes profile on Ollama. Do not change the hanshermesagent model identifier. Do not evict the 2B deriver or nomic-embed-text. Do not raise vLLM utilization. Do not add gpt-oss:120b. Treat tax math as code, not generation.

2. What HVE-CFO is for

HVE-CFO is the local financial analyst for Human Value Exchange as an Ontario CCPC and for the dual US-Canadian family that owns it. It classifies mixed personal and business spend, reads receipts, retrieves CRA / Ontario / IRS rules from a private corpus, drafts allocation memos, and supports treasury reporting. It does not file returns, does not invent tax payable, and does not post journal entries without a human or a deterministic ledger engine.

Job	Owner	Engine	Model	Why
Orchestrate, plan, draft memos, vision on receipts	hve-cfo	Ollama :11434	qwen3.8-hermes:27b-128k	Approved daily brain. Multimodal. 128K. Do not substitute qwen3.8:27b.
High-volume line classify / titles / compression	hve-cfo aux	Ollama :11434	qwen3.8-distill-2b:q4_k_m	Approved deriver. Keep resident. Cheap CSV passes.
RAG over CRA, IRS, chart of accounts, policy	hve-cfo	Ollama :11434	nomic-embed-text	Approved embedding model. Must stay hot.
Ledger scripts, parsers, tool patches	hve-coder (borrow)	vLLM :8000	Nemotron 3.5 Lightning only	Coder owns Lightning. CFO may request code via isolated MCP, never load Lightning itself.
Tax math, SBD headroom, % splits, GST	Deterministic layer	Host Python / spreadsheet / tax engine	—	LLMs draft. Calculators compute. Always.
Final tax position / filing	Human CA + Hans	—	—	Gray-area and filing authority stay human.

Table 1. One job, one owner. Lightning is not a second CFO brain.

3. Binding decisions inherited from the coder record

Luna’s 2 September 2026 decision record is the visible operational source of truth. HVE-CFO inherits every item that touches the shared box.

#	Approved decision	HVE-CFO implication
1	Preserve hanshermesagent model tag qwen3.8-hermes:27b-128k. The proposal's qwen3.8:27b is treated as incorrect.	CFO default model is the same exact tag. No generic 27B pull.

#	Approved decision	HVE-CFO implication
2	Preserve the three-model Ollama hot stack. Do not adopt OLLAMA_MAX_LOADED_MODELS=1.	CFO uses all three. It does not evict the deriver or embeddings to make room for another LLM.
3	While Ollama is resident, vLLM gpu-memory-utilization = 0.45. Fixed. No silent raise.	CFO load must fit inside the Ollama slice. No extra 60–75 GB model.
4	Nemotron Lightning is the complete coder backend. Coder will not load a second LLM.	CFO will not load Lightning in Ollama. Code work is delegated to hve-coder or written by deterministic scripts.
5	Persistent vLLM is a native venv + systemd user service. Docker is not the persistent method.	CFO does not introduce containers. Native Linux only, consistent with Hermes CFO master spec.
6	Strict workspace, path allowlists, MCP allowlists. No cross-profile writable access.	hve-cfo gets its own home, cwd, skills, and private books tree.
7–8	Automatic cloud fallback is intended, sanitized data only, provider still open.	CFO cloud fallback is off by default. Bank CSVs, SINS, account numbers, and tax slips never leave the box.
9–10	Promotion gate before retiring prior coder path. Start evaluation in dual-load.	hve-cfo is staged the same way. May 2026 four-model collective is not deleted until a CFO promotion gate passes.

Table 2. Decision inheritance. Source: agent-communications/2026-09-02-hve-coder-dual-backend-decisions-v1.0.md.

4. Models going forward — and models that are out

The May 2026 Hermes CFO master spec named a four-agent collective: Qwen 2.5 14B, Mistral Small 24B, Nemotron-3 Nano 30B-A3B, and Gemma 2 27B. That model set is superseded for inference. The roles (orchestrator, research, execution, critic) remain useful as prompt and skill roles inside one 27B brain plus the 2B deriver. They are no longer four resident Ollama weights.

Status	Tag / component	Role
IN — hot	qwen3.8-hermes:27b-128k	CFO orchestrator, memo writer, receipt vision, critic pass
IN — hot	qwen3.8-distill-2b:q4_k_m	Transaction classify, titles, compression
IN — hot	nomic-embed-text	Local RAG embeddings
IN — shared, not CFO-owned	Nemotron 3.5 Lightning on native vLLM :8000	Only via hve-coder isolation if a ledger tool must be written
OUT	gpt-oss:120b and gpt-oss:20b	Explicitly rejected for this stack going forward
OUT as hot residents	qwen2.5:14b, mistral-small:24b, nemotron-3-nano:30b, gemma2:27b	Superseded May 2026 collective weights
OUT as hot residents	qwen3.5:122b, llama3.3:70b, nemotron-3-super:120b, generic qwen3.8:27b	Do not preload. Evict if found.

Table 3. Forward model policy. Matches hermes-preload-models.sh evict list plus the 2 September decisions.

5. System layout

Three Hermes profiles share one GB10. They do not share memory stores, skills, sessions, or writable trees. They share one 128 GB unified LPDDR5X pool. Engine isolation is configuration. Memory is the real constraint.

Layer	hanshermesagent	hve-cfo	hve-coder
Home	~/hermes/	~/hermes/profiles/hve-cfo/	~/hermes/profiles/hve-coder/
Endpoint	Ollama 127.0.0.1:11434/v1	Same Ollama endpoint, different profile	Native vLLM 127.0.0.1:8000/v1
Default model	qwen3.8-hermes:27b-128k	qwen3.8-hermes:27b-128k	nemotron-3.5-lightning

Layer	hanshermesagent	hve-cfo	hve-coder
Auxiliary	2B deriver + nomic-embed-text	Same two aux models; CFO-specific prompts	Lightning only — no second LLM
Workspace	General HVE work	Dedicated books / tax / treasury cwd	Dedicated coder cwd
Writable allowlist	General HVE trees	books, receipts, tax-corpus, treasury logs only	coder workspace only
MCP / tools	General HVE tools	ledger, CSV, RAG, receipt ingest, report emit — no shell into coder or private runtime	git / test / patch tools — no books tree

Table 4. Profile isolation. A dedicated terminal.cwd is not sufficient isolation by itself (Decision 6).

NVIDIA DGX SPARK · GB10 · 128 GB unified LPDDR5X · one GPU
Ollama :11434 holds the three-model hot stack for hanshermesagent and hve-cfo.
vLLM :8000 holds Lightning for hve-coder only, utilization fixed at 0.45 while Ollama is resident.
Deterministic CFO tools (CSV parsers, allocation math, hledger/SQLite) run on CPU + disk and do not need a second LLM engine.
Watch `free -h`, not `nvidia-smi`.

6. Memory architecture on GB10

Unified memory is one LPDDR5X pool. vLLM pre-allocates KV from whatever it believes is free at start. The approved dual-load cap exists so the three Ollama weights plus a useful 32–128K Qwen context still fit. A 120B-class resident model was considered and rejected: it consumes the same slice Lightning needs and the same slice the 27B needs for long tax documents.

Slice	Dual-load target (default)	CFO-heavy day (coder idle)
OS + three Hermes gateways + sandboxes	~16–22 GB	~16–22 GB
Ollama 27B-128k + 2B + embed + 32–128K ctx	~28–40 GB	~28–48 GB (longer RAG context)
vLLM Lightning NVFP4 + DSpark + KV	~40–50 GB hard cap at 0.45	May stay warm; do not raise util
CFO SQLite / hledger / RAG index	~1–3 GB	~1–4 GB
Required headroom	≥20 GB	≥15 GB
Fourth hot LLM (gpt-oss:120b etc.)	Forbidden	Forbidden

Table 5. Unified-memory budget. Dual-load remains the default operating mode (Decision 10).

Operating rule. Start and stay at vLLM 0.45 while the Ollama stack is resident. Any increase requires a separate approved decision backed by telemetry. CFO-only burst means stopping vLLM, not loading a 120B model.

7. Profile and endpoint wiring

```
HVE-CFO profile — Ollama (approved tags)

model.default: qwen3.8-hermes:27b-128k
provider: custom
base_url: http://127.0.0.1:11434/v1
aux.deriver: qwen3.8-distill-2b:q4_k_m
aux.embed: nomic-embed-text
terminal.cwd: /home/.../hve-cfo-workspace
fs.allow: hve-cfo-workspace, tax-corpus, books-db
mcp.allow: cfo-ledger, cfo-rag, cfo-receipts, cfo-reports
```

Do not point hve-cfo at :8000. Do not set OLLAMA_MAX_LOADED_MODELS=1. Keep-alive on all three Ollama tags remains -1 / preload-service policy. If a ledger parser must be generated, emit a ticket to hve-coder; do not pull Lightning

into Ollama.

8. Cognitive roles inside one brain

The May 2026 four-seat design is retained as a prompt protocol, not as four GPUs’ worth of weights. Qwen 3.8 Hermes plays orchestrator, research synthesizer, and critic in separate turns with separate system prompts. The 2B deriver plays the cheap execution clerk for categorization. Deterministic code plays the calculator that Nemotron-Nano used to impersonate.

Seat	Was (May 2026)	Now	Hard rule
Orchestrator	Qwen 2.5 14B	qwen3.8-hermes:27b-128k	Only this tag routes CFO work.
Research	Mistral Small 24B	Same 27B + RAG corpus	Answers only from retrieved CRA/IRS/policy chunks plus books.
Execution clerk	Nemotron-3 Nano	2B deriver + Python tools	Classify and extract. Do not compute tax.
Critic / veto	Gemma 2 27B	Second 27B pass, critic prompt	Veto still exists for treasury/live-trade paths. Log it.
Code author	Nano / various	hve-coder + Lightning	CFO does not author production parsers in-session if coder is available.

Table 6. Role map. Weights collapsed; governance did not.

9. CFO workload design

Local models are good at classification, drafting, and citing a publication you uploaded. They are bad at tax calculations, current rates unless RAG’d, and positions you would defend in a CRA or IRS review. The architecture encodes that limit.

Loop	Input	Who runs it	Output
Weekly classify	Bank / card CSV	2B deriver → 27B review	Table: date, payee, amount, account, business %, personal %, benefit risk, source note
Receipt pass	Photo or PDF	27B vision	Vendor, GST/HST, purpose, mixed-use flag — no total invented
Rule question	Plain English + corpus	nomic RAG + 27B	Answer with cited chunk from T4002 / T2125 / Pub 535 / HVE policy
Quarterly memo	P&L + draws + family wages	27B + critic pass	SBD headroom questions, salary vs dividend questions, s.15(1) clean-up list — for the CA
Treasury report	Kraken / books totals	Deterministic totals + 27B prose	Calm Telegram voice from the v2.0 master spec. Numbers from code.

Table 7. Recurring CFO jobs. Family-business mixed-use is a first-class category, not an afterthought.

Standing system law for the 27B: You are a private bookkeeping analyst for an Ontario CCPC owned by a dual US-Canadian family. You do not file returns or invent tax amounts. Flag groceries, family travel, clothing, personal cards on the corp, commuting, mixed vehicle, home utilities without a floor-plan %, and family wages with no role description. Never compute tax payable.

10. Local tax corpus (RAG)

Training data is stale. The PDF is not. Load only what the current engagement needs. Embed with nomic-embed-text. Cite the passage.

Bucket	Examples	Use
Canada / Ontario	T4002, T2125 guide, home office, automobile, meals, s.15(1) shareholder benefit, TOSI, SBD \$500k ABI, GST/HST ITC	Default corpus for HVE CCPC

Bucket	Examples	Use
United States	IRS Pubs 17, 535, 463; HSA / HDHP notes relevant to citizenship	Overlay only when the question is US-side
HVE policy	Chart of accounts with Family / Personal / Corp / Mixed tags; salary policy; shareholder-loan policy; home-office % method	Highest-value retrieval
Optional skills pack	OpenAccountants Canada + ca-on + US markdown, stored offline	Current-rule text without a cloud MCP

Table 8. Corpus. Do not dump the entire CRA site into one index.

11. Operating modes

Mode	When	Actions
A — Dual resident	Normal HVE day: chat + books + occasional code	Ollama three-stack hot. vLLM at 0.45. hve-cfo on Ollama.
B — Coder burst	Focused implementation (already defined)	ollama stack may stay; do not raise vLLM. CFO defers heavy RAG.
C — CFO close	Month-end / year-end memo + receipt batch	Keep three Ollama models. Optionally stop vLLM to reclaim ~40 GB for 128K tax context.
D — Overnight agent	CSV batch + RAG ingest	Mode C. Cap parallel slots. Log classify confidence. No live trade, no filing.

Table 9. Default is A. Close week is C. There is no Mode E that loads gpt-oss:120b.

12. Risks and controls

Risk	Signal	Control
Fourth model eats the pool	ollama ps shows 120B / 70B / 122B	Preload script evict list. Refuse in profile config.
vLLM starves Qwen context	free -h collapses; long RAG stalls	Keep utilization at 0.45. Stop vLLM for Mode C.
Hallucinated tax number	Memo contains payable or rate not in corpus	Strip numbers from LLM output; inject calculator results only.
Shareholder-benefit miss	Personal spend coded 100% business	Critic pass + explicit mixed-use taxonomy + human review.
Cloud fallback leak	CSV or slip in a cloud prompt	CFO fallback off. If later enabled, sanitized metadata only (Decisions 7–8).
Workspace collision	CFO writes into coder or ~/hermes daily	Allowlists. No shared writable project root.
Old four-model spec revived	Someone pulls Gemma / Mistral / Nano hot	This document + preload policy supersede May 2026 weights.

Table 10. Failure modes on a single GB10 running three profiles.

13. Promotion gate

hve-cfo may not replace the prior Hermes CFO operating path, and may not be declared the production books agent, until all of the following exist:

#	Evidence required
1	Fixed classification benchmark on a held-out family/business CSV (precision on Mixed and Benefit-risk labels).
2	Receipt extraction sample: vendor / tax / amount digit accuracy, human-scored.
3	RAG citation test: answers to T4002 / home-office / s.15(1) questions include the correct source passage.
4	Zero invented tax-payable figures on a 20-question calc trap set.
5	Cold and warm p50 / p95 latency for classify and memo jobs under dual-load (vLLM at 0.45).

#	Evidence required
6	Mixed-load memory stability: free -h during a 20-minute CFO + coder session.
7	Workspace and MCP isolation validation (Decision 6).
8	Tested rollback to the prior CFO path and model tags.

Table 11. No implementation in this PDF is self-authorizing. Same bar the coder record set.

14. Rollout

#	Step	Done when
1	Confirm preload policy: three Ollama tags hot; gpt-oss:* and superseded collective evicted.	ollama ps shows only the approved three.
2	Create ~/.hermes/profiles/hve-cfo/ with SOUL, skills, allowlists, cwd.	Profile boots against :11434 with the hermes 27B tag.
3	Stand up local tax-corpus + nomic index (CRA/Ontario + HVE policy first).	One cited T4002 answer in-session.
4	Wire deterministic CSV + allocation tools. Ban LLM arithmetic in prompts.	A 50-line CSV returns percentages from code.
5	20-minute mixed load: CFO classify + coder tool loop with vLLM at 0.45.	free -h stable; neither engine restarts.
6	Run the promotion-gate suite. Keep May spec path intact until it passes.	Written results in agent-communications.

Table 12. Sequence. Do not skip the mixed-load hour.

15. What this is not

This is not a new inference engine. It is not permission to run gpt-oss:120b as a “reasoning CFO.” It is not a revival of the four hot Ollama weights from May 2026. It is not tax advice and not an e-filer. It is not Perplexity Portable Computer. It is not a two-Spark cluster plan. Until a second Spark exists, the 0.45 cap and the three-model Ollama stack *are* the architecture.

One-line summary. hve-cfo → qwen3.8-hermes:27b-128k + 2B deriver + nomic-embed-text on Ollama :11434. hve-coder → Lightning on native vLLM :8000 at 0.45. hanshermesagent keeps its exact model tag. Separate profile homes and cwds. No gpt-oss:120b. Calculators own the numbers.

Document version 1.0 · 2 September 2026 · Hardware assumed: 1× DGX Spark GB10, 128 GB unified memory. Incorporates Luna dual-backend decisions v1.0. Supersedes May 2026 CFO collective weights for inference only. Treasury risk rules, veto protocol, and Telegram voice from Hermes CFO Master Spec v2.0 remain in force.